Edward Lin

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Professional Experience

Software Engineer Sept 2026 – Present
Google – Cloud AI, Accelerator Software (ASIC-SW) *Taipei, Taiwan (Hybrid)*

- Develop the C++ golden reference model (GRM) for Google's Tensor Processing Unit (TPU) AI accelerators, a functional software simulator of the hardware used as the reference for validating silicon behavior.
- Work across the accelerator software toolchain supporting TPU architecture and verification teams.

Software Engineer Nov 2023 – Aug 2026
DAS Inc. – Developer of DAS Trader *Markham, Ontario, Canada*

- Independently designed and built a GPU-accelerated rendering engine from scratch using C++, OpenGL, and GLSL, replacing a 20-year-old CPU-based system, achieving a 15× increase in rendering speed, 42% reduction in CPU utilization, and enabling real-time visualization of high-frequency trading data.
- Leveraged RenderDoc for frame capture, shader debugging, and GPU performance analysis, identifying bottlenecks and optimizing render passes for maximum efficiency.
- Developed multi-GPU support with efficient context management, ensuring low-latency rendering across multiple GPUs and monitors for enhanced scalability.
- Introduced multi-threaded and parallel rendering from the ground up, enabling simultaneous real-time rendering of multiple stock charts with consistent frame rates under heavy data loads.
- Optimized real-time stock data processing by designing preprocessing logic from scratch, achieving 130K+ updates per second and reducing end-to-end latency by 26%.
- Collaborated directly with users to translate real-world workflow feedback into visualization and usability improvements.

Software Engineer Intern May 2021 – Apr 2022
Oracle Corporation *Toronto, Ontario, Canada*

- Contributed to the backend development of Oracle Responsys, working closely with the team on a large-scale codebase that powers a leading cross-channel marketing automation platform.
- Developed and optimized Java-based microservices using the Spring Framework, collaborating on 6+ new features that improved the efficiency of filtering millions of emails and SMS messages.
- Improved the data filtering algorithm and restructured data handling in existing code, reducing processing bottlenecks and achieving a 23% speed increase for large dataset operations.
- Enhanced backend server reliability by increasing unit test coverage by 15%, using JUnit to improve code maintainability and minimize regression issues.
- Resolved 12+ high-priority bugs that caused filtering algorithms to perform unexpectedly and slow with large datasets, improving backend reliability and optimizing system performance.

Skills

Programming Languages: C++, C, Java, Python, JavaScript, R, Shell Scripting
Frameworks & Libraries: React Native, React.js, Node.js, Django, PyTorch, TensorFlow, OpenCV, OpenMP
Graphics & GPU Computing: OpenGL, GLSL, RenderDoc
Tools & Databases: Git, SQL, JUnit

Education

University of Toronto Toronto, Ontario, Canada
Honours Bachelor of Science in Computer Science with a Statistical Minor *Sept 2018 – Nov 2023*

- Associate Programs: Focus in Artificial Intelligence, Focus in Web and Internet Technologies

Projects

Ontario Parks | Mobile Application github.com/Edward9292/ontario-parks-app

- Collaborated with Ontario Parks to design and develop a mobile audio-tour app using React Native and Node.js.
- Implemented user authentication, cross-device data synchronization, and Google Maps API integration with real-time location tracking for context-aware audio content delivery.

CartoonGAN | Machine Learning github.com/Edward9292/CartoonGAN

- Developed a Generative Adversarial Network (GAN) using Python, PyTorch, OpenCV, and NumPy to transform real images into cartoon-style visuals.
- Fine-tuned the GAN architecture and hyperparameters, reducing training time by 18% while maintaining high visual quality in generated images.